

Influencer Fraud Detection

Threshold Selection & Business Decision Framework (v2)

Prepared: 2026-08-12 | Production model: supervised rf plus anomaly flag | Validation: 800 labeled profiles (400 tune / 400 held-out test)

1. Executive Summary

Two approaches were tested head-to-head on an unseen test set: an unsupervised anomaly ensemble and a supervised classifier trained on labeled examples. The supervised model won decisively (precision 1.00, recall 0.98 vs. the ensemble's 0.50 / 0.20) and is now the production model. This report explains the resulting decision threshold and the operating trade-offs for the campaign approval team. All figures below are pulled from the same validation run referenced in the technical report -- there is a single number for the threshold, not two.

2. What Each Error Actually Costs the Business

Error Type	What Happens	Business Cost
False Positive (flagged, actually genuine)	A real, legitimate influencer is rejected or delayed for a campaign.	Lost partner, reputational friction with creators, wasted vetting overhead.
False Negative (missed, actually fraudulent)	A fake-follower or bot-driven account is approved and paid for a campaign.	Wasted ad spend, inflated reported reach, client trust and legal exposure.

For an influencer marketing agency, false negatives are typically the costlier error. This is why, once the production model cleared the 85% precision floor comfortably (1.00), the decision threshold below is set to favor recall rather than squeezing precision further.

3. Recommended Operating Threshold

Recommended fraud probability threshold: 50 / 100. This is the threshold actually used to produce predicted_fraud in the accompanying scored CSV and the technical report -- both documents are generated from the same pipeline run, so this number cannot drift out of sync with the technical validation again.

4. Resulting Vetting Action Breakdown (all profiles)

Vetting Action	Count
APPROVE - Low fraud risk	4380
REJECT - High fraud risk	555
MANUAL REVIEW - Audit history	42
CAUTION - Monitor 2-4 weeks	23

5. Choosing a Threshold by Business Priority

Business Priority	Threshold Direction	When to Use
Minimize wrongly rejecting good creators	Raise threshold (e.g. 65-80)	High-value, well-known creators; relationships the agency wants to protect.
Minimize fraud slipping through	Lower threshold (e.g. 35-45)	Large ad-spend campaigns, new/unfamiliar creator pools, strict brand-safety clients.
Balanced default	Keep at validated threshold (~50)	Standard campaigns without unusual risk or brand sensitivity.

6. Limitations & Governance

- The production model is trained on 800 labeled profiles. Fraud tactics evolve; this labeled set should be refreshed and the model re-validated on a recurring basis (e.g. quarterly).
- A supervised model can only recognize patterns similar to its training labels. The 'novel_pattern_flag' column (3 raised in this run) surfaces profiles the unsupervised anomaly detector considers unusual even though the supervised model scored them low-risk -- these deserve a manual look precisely because they don't resemble anything the model has seen labeled before.
- This is a decision-support tool, not an automated judge. Final rejection of an influencer, especially one with public brand reputation, should retain a human-in-the-loop step.
- If this pipeline is pointed at real data, replace data/influencers.csv with the actual raw export and confirm the 'is_fraud' column reflects verified outcomes, not proxy heuristics.